

BIM and Safety

How does Building Information Modeling create safer construction sites

Main Objective: BIM helps teams identify hazards, communicate risks in the field, and plan safer workflows



Prepared from your updated slide content

Topic: BIM and Safety

Essential Questions

Quiz at the end — pay attention.

1

What does BIM stand for?

2

Name one BIM software that is used to detect clashes.

3

What is the 4th dimension in Building Information Modeling?

4

How does BIM help eliminate language barriers?

Tip: all four answers appear again at the end.

What is BIM?

Building Information Modeling

An intelligent, 3D model-based process that provides architecture, engineering, and construction (AEC) professionals the insights and tools to efficiently plan, design, construct, and manage buildings and infrastructure.

More than a Model

Adds data about materials, costs, and lifecycles into models to improve collaboration and reduce errors.

Models are not new. For example, Sir Christopher Wren's Great Model was completed in 1672. (Picture) (Source: Autodesk)

Ordinary 3D models are becoming obsolete with BIM software. The added data and collaboration are creating safer work environments.

3D

geometry and visualization

Data

materials, cost, and lifecycle info

Collaboration

shared information helps architects, engineers, contractors, and owners make better decisions

Why it matters for safety

When geometry, schedule, and jobsite information are connected, teams can see hazards earlier and coordinate safer work.

Why BIM matters

Construction is still high risk, so prevention has to start before work begins.

1 in 5

workplace deaths occurred
in construction in 2023

38.5%

of construction deaths were
due to falls, slips, and trips

The best value of BIM is that it moves upstream — into design, planning, and coordination — instead of waiting until people are already exposed on site.

OSHA Focus Four hazards

Falls

Struck-by

**Caught-in /
in between**

Electrocution

If those are the recurring hazards, a better model of the work can help teams see them earlier and add controls sooner.

What does BIM add to Safety

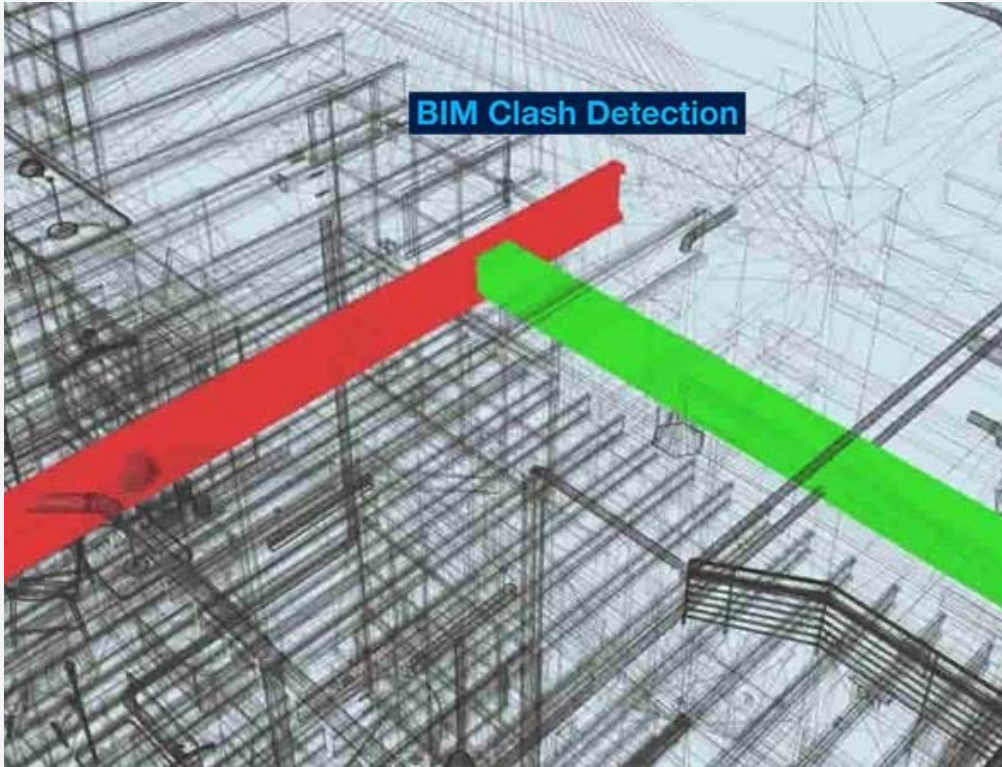
BIM becomes valuable when geometry, time, data, and people are connected.



Bottom Line: Safer planning and site decisions

Designs out hazards before construction starts

The safest fix is the one made before crews are exposed in the field.



Navisworks: Design software used to review and detect clashes, simulate 4D/5D schedules, and analyze projects. It combines multiple file formats (RVT, DWG, etc.) to identify interferences prior to construction.

Clash Detection can add safer details earlier:

- Guardrails
- Parapets
- Access routes
- Safer maintenance space

NIOSH

Designing work to remove hazards is the most reliable and effective way to protect workers.

Sources: Zhang et al. (2013), Autodesk University (2017), NIOSH PtD Toolkit (2024)

BIM has 4D not just 3D

What is 4D and how is it useful? Time.

**4D BIM = model +
schedule**

What this adds: it can monitor site logistics as construction happens — deliveries, access routes, crane locations, worker routes, and emergency egress routes.

**Structure
open edges**

**Envelope
scaffolds**

**MEP rough-in
trade overlap**

**Interiors
material flow**

**Commissioning
access + handoff**

This helps with building life cycle and can allow FMs to see emissions and equipment maintenance.

Proprietary objects: digital 3D representations of real-world products containing detailed geometric, technical, and parametric data.

Improving communication, training, and coordination

A model is often easier to understand than a long written instruction.

Models are easier to understand than long written instructions. BIM helps communicate visually and helps eliminate language barriers.

- Visualize safest sequence of work during huddles
- Reduced misunderstandings across different trades and language backgrounds
- Field use matters. BIM only improves safety if the model is used by people who are planning and supervising the work

Autodesk note

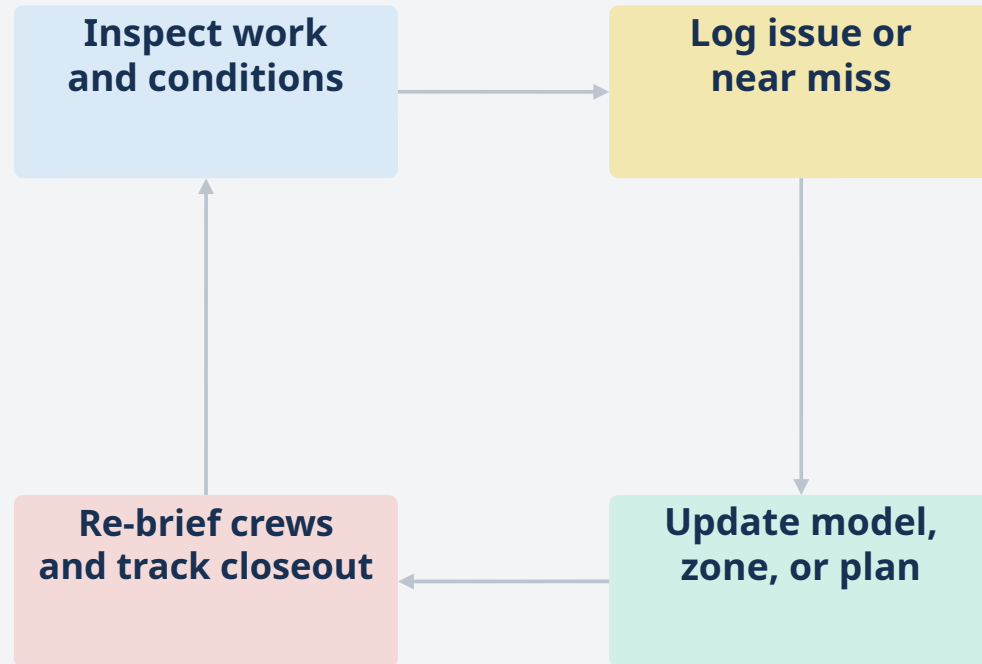
BIM's visual communication can break down language barriers and help workers understand how the asset is best constructed.



Field use matters: BIM only improves safety if the model is actually used by the people planning and supervising the work.

Turn inspections and near misses into feedback

Near miss data can reveal patterns before an injury event occurs.



Issue tracking becomes more useful when it is tied to a location in the model.

Near-miss data can reveal patterns before an injury event occurs.

The value is not just detection — it is learning and changing the next plan.

CDC/CPWR example

Near-miss reporting data was used to create near-miss locations in BIM so teams could mitigate hazards before an injury event.

How BIM improves safety on a real project

The same content from your slide, reformatted into a cleaner 5-step layout.

1

Plan for safety in BIM

Decide what safety items will be modeled, who is responsible, and when they will be reviewed.

2

Model the risky areas

Include openings, edges, crane paths, access routes, laydown areas, and temporary works.

3

Review dangerous work early

Use coordination meetings and 4D scheduling to look at high-risk tasks before crews start.

4

Use the model in the field

Bring BIM into pre-task plans, safety meetings, inspections, and jobsite walks.

5

Train people and hand it off

Give the owner and field team usable safety information and make sure they know how to use it.

Challenges/Benefits (2025)

Bottom Line

BIM makes job sites safer when teams plan hazards early, review risky work before it happens, and use the model during construction instead of just creating it and forgetting about it.

Plan hazards early

Review risky work before it happens

Use the model during construction

Do not just create it and forget it

QUIZ

What does BIM stand for?

Building Information Modeling

Name one BIM software that is used to detect clashes?

Navisworks

What is the 4th dimension in Building Information Modeling?

Time

How does BIM help eliminate language barriers?

Visual aid

Sources

References retained from your updated presentation.

Zhang, S., Teizer, J., Lee, J.-K., Eastman, C. M., & Venugopal, M. (2013). Building Information Modeling (BIM) and Safety: Automatic Safety Checking of Construction Models and Schedules. *Automation in Construction*, 29, 183-195.

<https://doi.org/10.1016/j.autcon.2012.05.006>

Autodesk University. (2017). BIM for Health and Safety in Construction. <https://www.autodesk.com/autodesk-university/article/BIM-Health-and-Safety-Construction-2017>

Alotaibi, A., & Gambatese, J. A. (2025). Challenges and Benefits of Using BIM Technologies to Improve Construction Safety: An Exploratory Case Study. *CIB Conferences*, 1, Article 393.

<https://doi.org/10.7771/3067-4883.1028>

BIMSafe NZ. (2024). BIM for Health and Safety: Construction Phase Report. https://guidance.bimsafe.nz/wp-content/uploads/2024/07/BIMSafe-Report_Construction_final.pdf

SafeWork NSW. WHS Management using Building Information Modelling (BIM) - Phase 2 Report. <https://www.safework.nsw.gov.au/resource-library/whs-research/BIM-phase-2-report.pdf>

U.S. Bureau of Labor Statistics. (2025). Fatal falls in the construction industry in 2023. <https://www.bls.gov/opub/ted/2025/fatal-falls-in-the-construction-industry-in-2023.htm>

CPWR / CDC. (2016). Integrating Safety Data into BIM Models Used in Project Design and Construction.

OSHA Focus Four Hazards
<https://www.osha.gov/training/outreach/construction/focus-four>